# AI and Ethics: Opportunities and Challenges

An Innovative Assignment for the Subject of Professional Ethics

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# The Duality of Artificial Intelligence

Artificial intelligence represents one of the most transformative technologies of our era, mimicking human problem-solving and decision-making capabilities. Its behavior is fundamentally shaped by the data it processes and the interactions it encounters, making every engagement with AI an inherently ethical one that demands careful consideration.

### Machine Learning

The foundational technology powering modern AI systems. Enables computers to learn patterns from data without explicit programming. System performance depends critically on training data quality and representativeness.

### Generative AI

Creates original, complex content including text, images, and audio. Poses ethical challenges around misinformation, intellectual property rights, and questions of authenticity in digital content.

### Agentic AI

Autonomous programs that accomplish goals without direct human interaction. Raises critical questions about control, accountability, and the potential for unintended consequences in automated systems.

Professional engineers stand at a critical crossroads, required to uphold ethical standards to ensure AI benefits society while actively minimizing its inherent risks and potential harms.

# Study Objectives and Research Methodology

## Critical Analysis of Ethical Challenges

- **Algorithmic Bias:** Investigating how human biases embedded in data and design lead to discriminatory outcomes
- **Data Privacy & Surveillance:** Examining risks of mass data collection and erosion of personal privacy
- **System Opacity & Accountability:** The "black box" problem and difficulty assigning responsibility when AI fails
- **Socio-Economic & Environmental Impacts:** Analyzing effects on employment, energy consumption, and resource usage

## Evaluation of Transformative Opportunities

- **Healthcare:** Improving disease detection, personalizing treatment plans, and accelerating medical research breakthroughs
- **Environmental Sustainability:** Enhancing climate models, optimizing renewable energy systems, and managing natural resources
- **Human Accessibility:** Developing technologies that empower individuals with disabilities and promote independence

**Research Methodology:** A qualitative analysis synthesizing previous research, prominent case studies, and public perception data to explore AI's dual nature as both opportunity and challenge.

# Case Study: Algorithmic Bias in Criminal Justice

## The COMPAS Recidivism Tool

### Background and Findings

The COMPAS algorithm is used throughout the U.S. justice system to predict the likelihood of a defendant re-offending. A groundbreaking 2016 ProPublica investigation revealed significant racial bias in its predictions, despite an overall accuracy of only 62%.

### False Positive Rate

Black defendants who did **not** re-offend were nearly twice as likely to be misclassified as "high-risk" compared to white defendants: 45% versus 23%.

### False Negative Rate

White defendants who **did** re-offend were more likely to be misclassified as "low-risk": 48% versus 28% for Black defendants.

**45%**

**Black defendants falsely labeled high-risk**

**23%**

**White defendants falsely labeled high-risk**

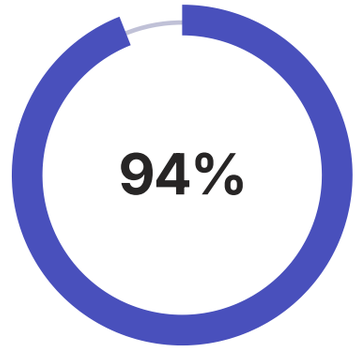
**62%**

**Overall algorithm accuracy**

**Critical Implication:** This case powerfully demonstrates how algorithmic tools, if not carefully designed, rigorously tested, and continuously audited, can perpetuate and even amplify existing societal inequalities, raising profound concerns about fairness and justice in automated decision-making systems.

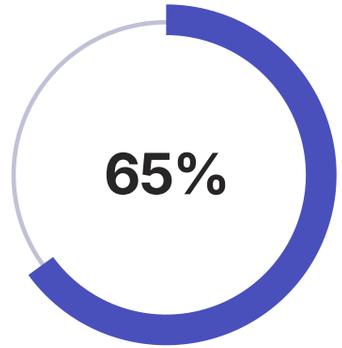
# AI for Good: Medical Breakthrough

## AI-Enhanced Lung Nodule Detection




**AI Model Accuracy**

Detecting early-stage lung cancer




**Human Radiologist Accuracy**

Traditional diagnostic methods

A groundbreaking model developed collaboratively by MIT and Massachusetts General Hospital achieved 94% accuracy in identifying early-stage lung nodules from CT scans—significantly outperforming the 65% accuracy rate of experienced human radiologists working independently.

> AI serves as a powerful assistive tool in medicine, enhancing human diagnostic capabilities to improve patient outcomes, reduce errors, and save lives through earlier detection of life-threatening conditions.

# Public Perception: A Tale of Two Nations

## United States: Cautious Concern

A comprehensive 2025 Pew Research survey revealed that **50% of Americans** are more concerned than excited about AI's growing role in society.

### Primary Concerns

- Weakening of essential human skills and critical thinking abilities
- Proliferation of misinformation and deepfake content
- Loss of human control over important decision-making processes
- Privacy violations and data security vulnerabilities

## India: Optimistic Adoption

Indian citizens demonstrate remarkable enthusiasm for AI, driven by tangible benefits experienced in daily life and rapid technological adoption across sectors.

### Key Drivers of Optimism

- **87% adoption rate** of AI-powered FinTech applications
- Sophisticated UPI fraud detection systems protecting millions of transactions
- Local language chatbots dramatically improving digital accessibility
- Personalized e-commerce experiences enhancing consumer satisfaction

# The Engineer's Ethical Mandate

## Key Finding: The Inherent Duality of AI

AI's greatest strengths are inextricably linked to its most significant ethical risks. The same capabilities that enable breakthrough innovations can, without careful governance, lead to serious harm.

### Data Analysis Power

Medical breakthroughs OR Biased decisions

### System Autonomy

Operational efficiency OR Accountability gaps

### Personalization

Enhanced convenience OR Privacy erosion

## Six Ethical Principles for AI Engineers

01

### Human-Centric Design

Prioritize human well-being, dignity, and maintain meaningful human control over critical systems

02

### Proactive Bias Mitigation

Actively audit data and models for fairness from inception through deployment and beyond

03

### Privacy & Security by Design

Embed robust data protection into the system's core architecture, not as an afterthought

04

### Transparency & Accountability

Strive for explainable systems and establish clear lines of responsibility for AI decisions

05

### Holistic Impact Assessment

Consider the full spectrum of social, environmental, and economic costs and benefits

06

### Lifelong Learning Commitment

Stay current on evolving technology, ethical frameworks, and relevant legal requirements

**The future of AI is not just about what we *can* build, but what we *should* build.**

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This presentation was prepared for the Professional Ethics course at Veer Surendra Sai University of Technology, Burla, demonstrating the critical intersection of engineering practice and ethical responsibility in the age of artificial intelligence.